

**EJADA EGYPT SUMMER INTERNSHIP PROGRAM 2026**

Cloud Build Track — OCI &amp; Terraform

**Week 3 — Lab 3**

# Terraform Implementation & Evidence Report

*Containerized NGINX Workload on Oracle Kubernetes Engine — VCN-Native Networking, a Managed Node Pool, and CSI-Backed Block Storage*

|                      |                                           |
|----------------------|-------------------------------------------|
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| <b>Region</b>        | me-jeddah-1                               |
| <b>Compartment</b>   | intern-18-salah-abdelhady-cmp             |
| <b>Working stack</b> | terraform/nonprd/week3-containerized-oke/ |

# Contents

|                                                  |           |
|--------------------------------------------------|-----------|
| <b>1. Objectives</b>                             | <b>3</b>  |
| <b>2. Environment</b>                            | <b>3</b>  |
| <b>3. Architecture</b>                           | <b>5</b>  |
| 3.1 Target architecture (Terraform)              | 5         |
| 3.2 HTTP traffic path                            | 5         |
| 3.3 Resource roles                               | 6         |
| <b>4. Source Inventory</b>                       | <b>7</b>  |
| 4.1 Root files                                   | 7         |
| 4.2 modules/network                              | 8         |
| 4.3 modules/subnet                               | 8         |
| 4.4 modules/oke                                  | 9         |
| 4.5 k8s/ application layer                       | 9         |
| <b>5. Variables and terraform.tfvars.example</b> | <b>10</b> |
| <b>6. Terraform Design Standards Applied</b>     | <b>11</b> |
| <b>7. Commands Executed (Successful Path)</b>    | <b>12</b> |
| 7.1 Tooling baseline                             | 12        |
| 7.2 Part 1 — network apply                       | 12        |
| 7.3 Part 2 — OKE apply                           | 12        |
| 7.4 Workload deployment via kubectl              | 12        |
| 7.5 Destroy                                      | 13        |
| <b>8. Evidence Catalogue</b>                     | <b>14</b> |
| <b>9. Screenshot Evidence</b>                    | <b>16</b> |
| 9.1 Stage A — network                            | 16        |
| 9.2 Stage B — OKE                                | 22        |
| 9.3 Stage C — workload and destroy               | 24        |
| <b>10. Issues and Troubleshooting</b>            | <b>27</b> |
| <b>11. Cleanup and Destroy Verification</b>      | <b>29</b> |
| <b>12. Document Index</b>                        | <b>30</b> |

## 1. Objectives

This lab rebuilds the Week 3 workload entirely through Terraform, without a manual Console-first pass. The stack provisions the network, the OKE cluster, and a containerized NGINX workload backed by block storage, then is destroyed cleanly once the evidence below was captured.

1. Deploy the Week 3 environment 100% with Terraform: a VCN on 10.2.0.0/16, three subnets, and an OKE cluster.
2. Network module — VCN, Internet Gateway, NAT Gateway, Service Gateway, and a shared log group (w3-vcn, w3-igw, w3-nat, w3-sgw, w3-vcn-logs).
3. Subnet module — subnet, route table, security list, and optional flow logs, instantiated three times for lb, workers, and pods.
4. OKE module — a cluster with VCN-native pod networking (OCI\_VCN\_IP\_NATIVE), a managed worker node pool, and dedicated API, worker, and pod NSGs.
5. Once the cluster reached ACTIVE, an NGINX Deployment, a CSI-backed block volume PVC (oci-bv, mounted at /data), and a Service of type LoadBalancer.
6. A clean destroy after evidence capture, leaving no billable resources behind.

### Building blocks delivered

| Building block                       | What landed                                                                                    |
|--------------------------------------|------------------------------------------------------------------------------------------------|
| Network related-group                | module.network — w3-vcn, Internet Gateway, NAT Gateway, Service Gateway, log group w3-vcn-logs |
| Subnet related-group                 | module.subnet × 3 (w3-lb-subnet, w3-workers-subnet, w3-pods-subnet)                            |
| OKE related-group                    | Cluster w3-oke ACTIVE, node pool w3-np (size 1), CNI OCI_VCN_IP_NATIVE                         |
| Application layer (PVC / Service LB) | kubectI-managed namespace w3-demo, PVC Bound 50Gi, Service VIP 193.122.68.107                  |
| Destroy                              | 36 Terraform resources destroyed; Kubernetes Service, PVC, and namespace removed first         |

## 2. Environment

| Item                        | Value                            |
|-----------------------------|----------------------------------|
| Tenancy                     | ociejada                         |
| Compartment                 | intern-18-salah-abdelhady-cmp    |
| Region                      | me-jeddah-1                      |
| Availability domain         | oXVt:ME-JEDDAH-1-AD-1            |
| Terraform                   | v1.15.8 (windows_amd64)          |
| Provider                    | oracle/oci v6.37.0 (~> 6.0)      |
| Authentication              | OCI config profile DEFAULT       |
| VCN                         | w3-vcn — 10.2.0.0/16             |
| Public load balancer subnet | w3-lb-subnet — 10.2.1.0/24       |
| Private workers subnet      | w3-workers-subnet — 10.2.10.0/24 |
| Private pods subnet         | w3-pods-subnet — 10.2.128.0/18   |

| Item                     | Value                                                     |
|--------------------------|-----------------------------------------------------------|
| Kubernetes services CIDR | 10.96.0.0/16 (outside the VCN)                            |
| Tags                     | Project=Ejada-Cloud-Build, Lab=week3, ManagedBy=Terraform |

### Runtime values captured during the lab

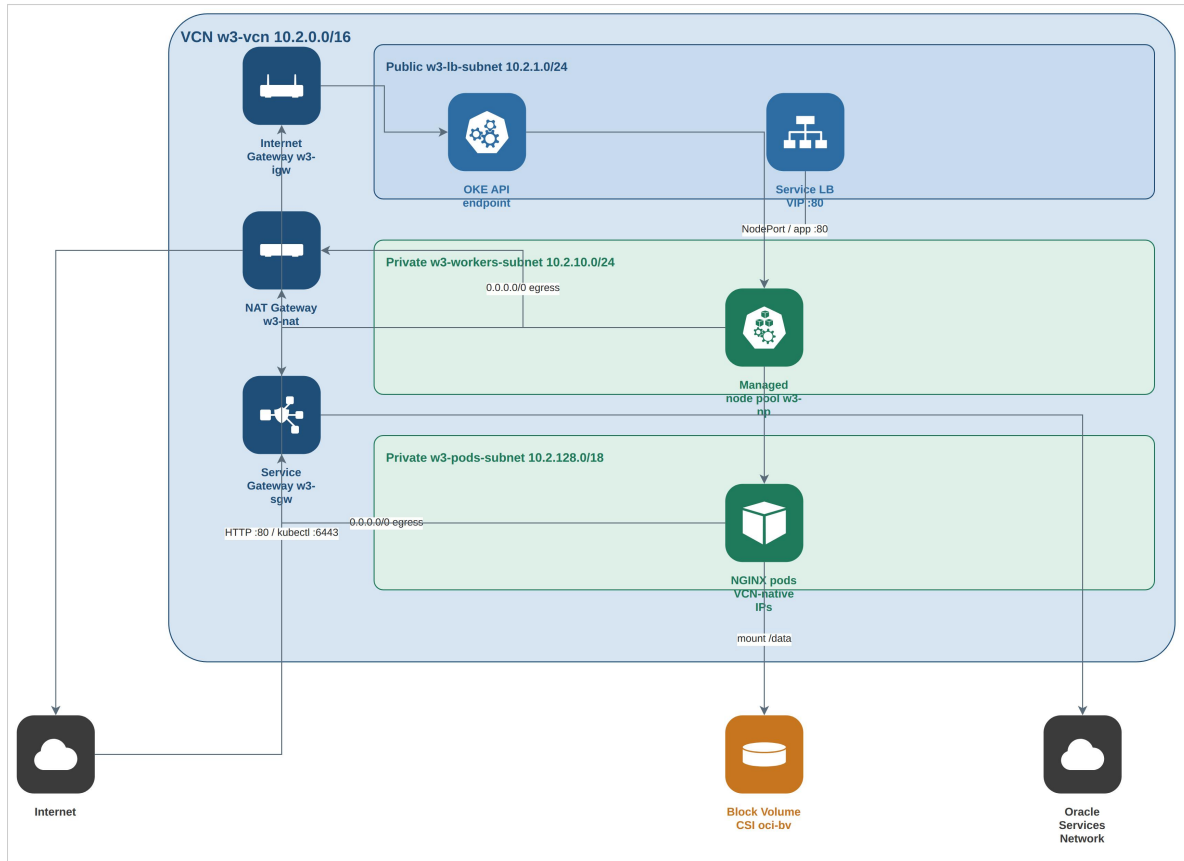
*The cluster and workload below were live for the duration of validation and were destroyed afterward; the values are retained here as evidence.*

| Item                  | Value                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------|
| Cluster               | w3-oke — ACTIVE, BASIC, Kubernetes v1.34.1, CNI OCI_VCN_IP_NATIVE                         |
| Cluster OCID          | ocid1.cluster.oc1.me-jeddah-1.aaaaaaaavhl7onzm5htuw5k2reau64kdc7effu4itsy5qyztsclhoyslgma |
| Node pool             | w3-np — size 1, VM.Standard.E4.Flex (1 OCPU / 8 GB)                                       |
| Worker                | Private IP 10.2.10.193, no public IP                                                      |
| Namespace             | w3-demo                                                                                   |
| PVC                   | nginx-data — Bound, 50Gi, StorageClass oci-bv                                             |
| Image                 | docker.io/library/nginx:alpine                                                            |
| Service load balancer | External IP 193.122.68.107                                                                |
| Volume mount          | /data on the NGINX container                                                              |

The Part 1 and Part 2 CLI snapshots reference different internal VCN OCIDs from the same apply-and-recreate cycle, though the display names and CIDRs are identical across both. The names, CIDRs, and cluster OCID recorded in this document are the values of record. No secrets, keys, or tokens are included.

## 3. Architecture

### 3.1 Target architecture (Terraform)



**Figure 1.** Target network architecture — VCN, gateways, subnets, and the OKE workload path.

The Internet-facing path runs through the Internet Gateway into the public load balancer subnet, which carries both the OKE API endpoint and the Kubernetes Service load balancer. Workers and pods stay fully private, with no public IP addresses assigned. The NAT Gateway covers general egress for workers and pods, while the Service Gateway reaches the Oracle Services Network for the CSI driver and other regional APIs. Pods take VCN-native IP addresses from the 10.2.128.0/18 pods subnet rather than an overlay CIDR.

### 3.2 HTTP traffic path



**Figure 2.** Request path from the browser through the Service load balancer to the pod, and the CSI volume mount.

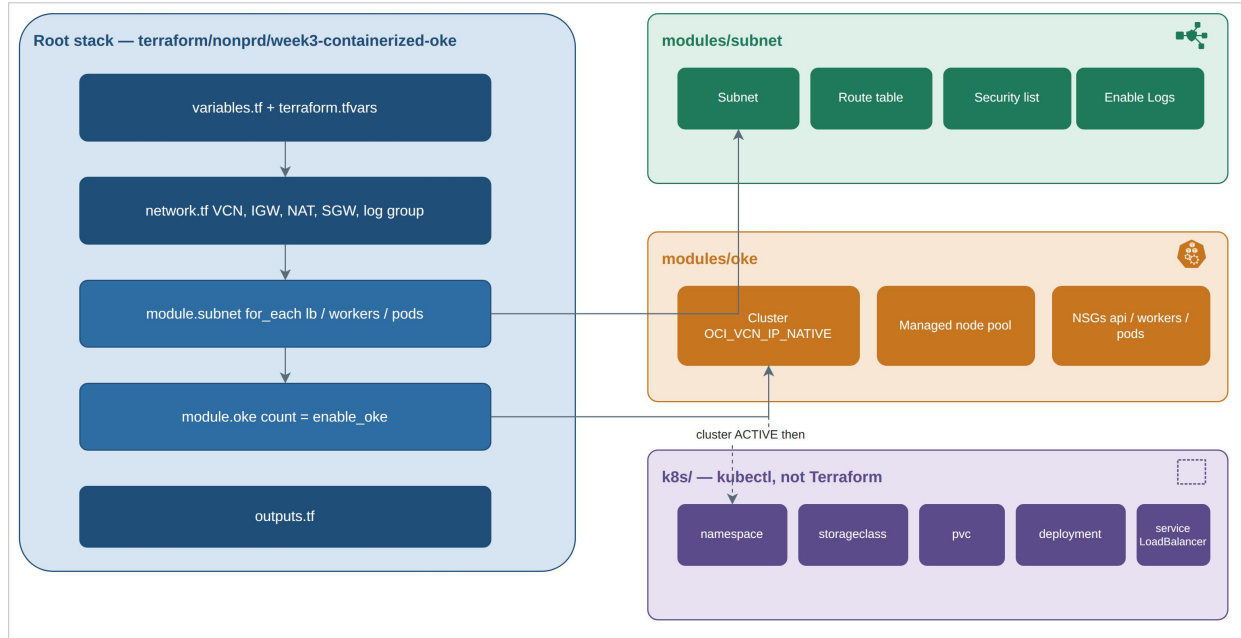
Clients never reach workers or pods directly. The Service of type LoadBalancer provisions an OCI Flexible Load Balancer in the public subnet, which forwards to the worker's NodePort and on to the container through kube-proxy. The persistent volume is an external Block Volume attached through the CSI driver (oci-bv), not File Storage. The historical VIP for this lab was 193.122.68.107, and the worker's private IP was 10.2.10.193.

### 3.3 Resource roles

| Layer                                         | Resources                                                                                            |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------|
| Network module (network.tf → modules/network) | VCN, Internet Gateway, NAT Gateway, Service Gateway, shared log group w3-vcn-logs                    |
| Subnet module × 3 (subnets.tf)                | Per key: subnet + route table + security list; the lb subnet also adds a flow log and capture filter |
| OKE module (oke.tf; NSGs in nsg.tf)           | Cluster, managed node pool, NSGs (w3-nsg-oke-api / -workers / -pods)                                 |
| Application layer (kubectl, not Terraform)    | Namespace, PVC on oci-bv, Deployment, Service load balancer                                          |

## 4. Source Inventory

The root module lives at `terraform/nonprd/week3-containerized-oke/` and calls three cohesive modules: `modules/network`, `modules/subnet`, and `modules/oke`. The recommended read order is `versions.tf` → `providers.tf` → `variables.tf` → `locals.tf` → `network.tf` → `subnets.tf` → `oke.tf` → `outputs.tf`. There is no `data.tf`: the All-OSN lookup lives inside `modules/network`, and the availability-domain and node-image catalogs are queried directly in `oke.tf`.



**Figure 3.** Module map — root stack calling the subnet and OKE modules, with the application layer applied afterward.

### 4.1 Root files

| File                                  | Purpose                                                                                                                                                                                                                                                                                      |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>versions.tf</code>              | Terraform >= 1.5; required provider oracle/oci ~> 6.0                                                                                                                                                                                                                                        |
| <code>providers.tf</code>             | OCI provider: region + config_file_profile; ignore_defined_tags for the intern-uneditable Oracle-Tags.CreatedBy / CreatedOn; optional API-key variables stay null under a CLI profile                                                                                                        |
| <code>variables.tf</code>             | All tunable arguments — compartment, CIDRs, names, shapes, OCPU/memory, ports, counts, tags, DNS labels, feature flags (enable_oke, enable_service_gateway, enable_oke_nsgs, enable_https_ingress), log retention, Kubernetes version, node image, SSH public key                            |
| <code>terraform.tfvars.example</code> | Committed template with placeholder values only; real values stay in the gitignored <code>terraform.tfvars</code>                                                                                                                                                                            |
| <code>locals.tf</code>                | Freeform tags; display names derived from the name prefix or overrides (w3-vcn, w3-oke, NSGs, ...); availability-domain and OKE version/image selection; security-list and route-table rule payloads; the subnets map passed into module.subnet with gateway IDs sourced from module.network |
| <code>network.tf</code>               | module.network (VCN, Internet Gateway, NAT Gateway, Service Gateway, shared log group); moved blocks map earlier root-level addresses into the module                                                                                                                                        |
| <code>subnets.tf</code>               | module.subnet for_each across lb, workers, and pods                                                                                                                                                                                                                                          |
| <code>oke.tf</code>                   | module.oke with count = var.enable_oke ? 1 : 0; availability-domain and node-image data sources; wires subnet IDs and NSG names into the module                                                                                                                                              |

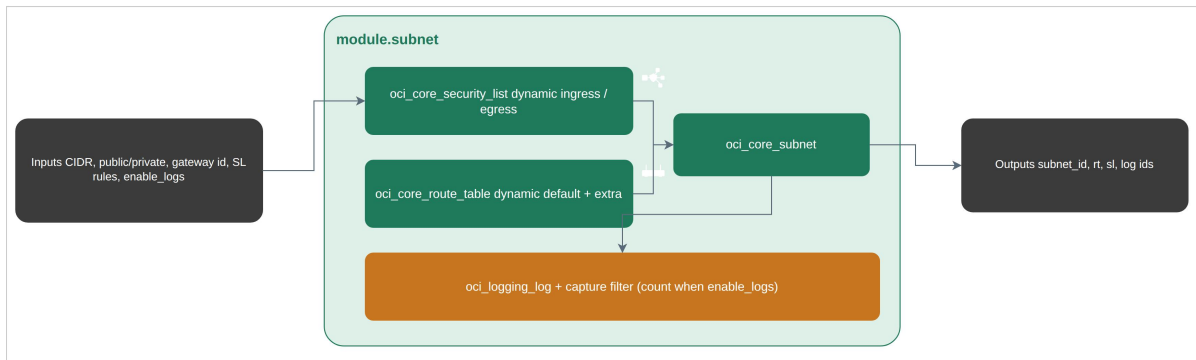
| File                | Purpose                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------|
| outputs.tf          | VCN and subnet IDs, Service Gateway, log group, cluster and node-pool IDs, CNI type, NSG IDs, and kubeconfig / kubectl-apply hints |
| .terraform.lock.hcl | Locked provider version 6.37.0                                                                                                     |
| README.md           | Stack operator notes; this document remains the authoritative narrative for the full lab                                           |

## 4.2 modules/network — VCN, gateways, shared log group

| File         | Purpose                                                                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| main.tf      | All-OSN data source; oci_core_vcn; Internet Gateway; NAT Gateway; Service Gateway (count when enable_service_gateway); shared oci_logging_log_group (count when any subnet has logging enabled) |
| variables.tf | Display names, VCN CIDR/DNS, Service Gateway and log-group flags, tags                                                                                                                          |
| outputs.tf   | vcn_id, gateway IDs, log_group_id, osn_cidr                                                                                                                                                     |
| versions.tf  | Inherits the root OCI provider                                                                                                                                                                  |

*This is one related group, not a module per gateway. Subnets stay in modules/subnet, and cluster NSGs stay in modules/oke.*

## 4.3 modules/subnet — subnet, route table, security list, logs



**Figure 4.** Subnet module internals — one related group instantiated per subnet key.

| File         | Purpose                                                                                                                                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| main.tf      | Security list (dynamic ingress/egress with nested TCP/UDP/ICMP options), route table (dynamic default 0.0.0.0/0 plus extra rules), subnet, and an optional capture filter and service flow log (count when enable_logs) |
| variables.tf | Display name, CIDR, DNS, public/private flag, gateway ID, extra routes, security-list rules, log flags (enable_logs, create_log_group, log_group_id)                                                                    |
| outputs.tf   | subnet_id, route-table and security-list IDs, log and capture-filter IDs                                                                                                                                                |
| versions.tf  | Inherits the root OCI provider                                                                                                                                                                                          |

For this lab, create\_log\_group is false and log\_group\_id is sourced from module.network. Only the lb subnet has enable\_logs = true. Workers and pods route their default 0.0.0.0/0 through NAT with a Service Gateway extra rule, while the lb subnet routes its default through the Internet Gateway.



## 4.4 modules/oke — cluster, VCN-native CNI, managed pool

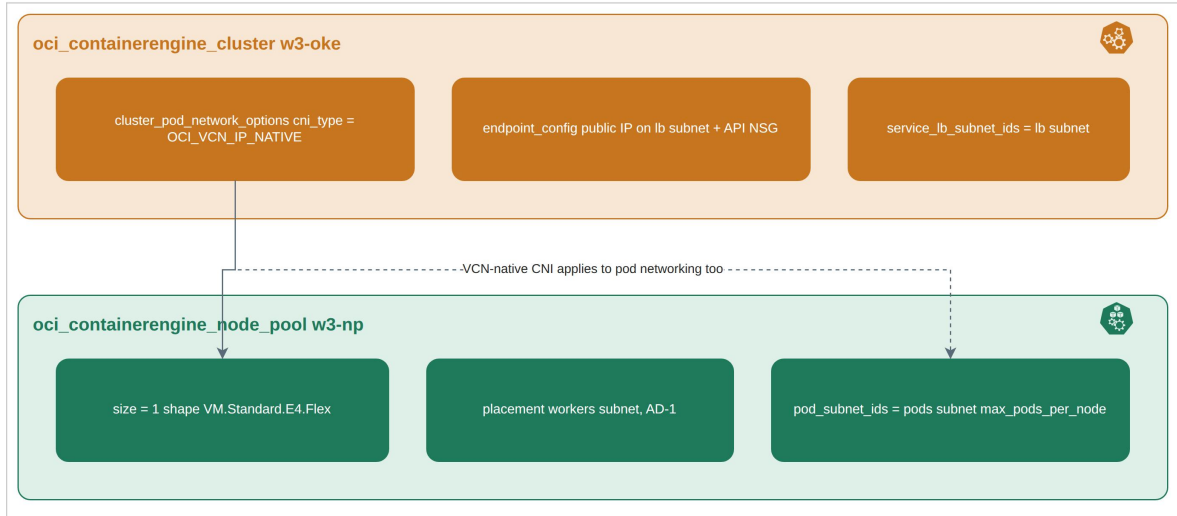


Figure 5. OKE cluster and node pool, both configured for VCN-native pod networking.

| File         | Purpose                                                                                                                                                           |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| main.tf      | oci_containerengine_cluster with cni_type = OCI_VCN_IP_NATIVE; oci_containerengine_node_pool with the worker subnet, pod subnet IDs, and a Flex node_shape_config |
| nsg.tf       | API, worker, and pod NSGs plus rules (count / for_each), attached to the API endpoint, workers, and VCN-native pod VNICS                                          |
| variables.tf | Cluster name/type/version/CNI, endpoint/worker/pod/load-balancer subnet IDs, node shape/OCPU/memory/size/image, SSH key, NSG names, CIDRs, and ports              |
| outputs.tf   | Cluster and node-pool IDs, endpoints, CNI type, NSG IDs                                                                                                           |
| versions.tf  | Inherits the root OCI provider                                                                                                                                    |

NSGs live inside this module rather than as a root file or a one-resource wrapper. Kubernetes workloads themselves are not a Terraform module.

## 4.5 k8s/ — application layer after the cluster exists

| File              | Purpose                                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| namespace.yaml    | Namespace w3-demo                                                                                                                                                                                |
| storageclass.yaml | Explicit oci-bv definition (blockvolume.csi.oraclecloud.com). The cluster already ships this class, so this manifest is kept for reference; see Section 10.2 for how a reapply of it was handled |
| pvc.yaml          | PVC nginx-data, 50Gi, ReadWriteOnce, storageClassName: oci-bv                                                                                                                                    |
| deployment.yaml   | NGINX Deployment, image docker.io/library/nginx:alpine, volume mount at /data                                                                                                                    |
| service.yaml      | Service type: LoadBalancer, Flexible shape 10–10 Mbps annotations                                                                                                                                |
| README.md         | kubectl apply / delete order for this layer                                                                                                                                                      |

Not committed: terraform.tfvars, terraform.tfstate\*, .terraform/, local plan binaries, and private keys.

## 5. Variables and terraform.tfvars.example

Changeable arguments live in variables.tf. Environment-specific values live in the gitignored terraform.tfvars; the committed template is terraform.tfvars.example.

| Variable                                   | Role in this lab                                                                                         |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------|
| compartment_ocid                           | intern-18 (placeholder value in the example file)                                                        |
| region / config_file_profile               | me-jeddah-1 / DEFAULT                                                                                    |
| name_prefix / project_tag / lab_tag        | w3 / Ejada-Cloud-Build / week3                                                                           |
| enable_oke                                 | false for the Part 1 network apply; true for the Part 2 cluster and node-pool apply                      |
| enable_service_gateway / enable_oke_nsgs   | both true                                                                                                |
| vcn_cidr / subnets                         | 10.2.0.0/16; keys lb, workers, pods with CIDRs, DNS labels, public/private flag, route_type, enable_logs |
| allowed_ssh_cidr                           | Laptop CIDR for API :6443 and worker SSH (placeholder x.x.x.x/32 in the example)                         |
| ssh_public_key                             | Node metadata; the example file shows a dummy ssh-ed25519 AAAA... line                                   |
| oke_cluster_type / oke_cni_type            | BASIC_CLUSTER / OCI_VCN_IP_NATIVE                                                                        |
| oke_endpoint_public                        | true, to allow kubectl from a laptop                                                                     |
| oke_services_cidr                          | 10.96.0.0/16                                                                                             |
| oke_node_pool_size / shape / OCPU / memory | 1 / VM.Standard.E4.Flex / 1 / 8 GB                                                                       |
| k8s_namespace                              | w3-demo                                                                                                  |

Display names such as w3-vcn, w3-oke, and the NSG names are derived in locals.tf from name\_prefix unless an override is set; root resources do not hardcode them.

### Example (placeholder values, not the live tfvars)

```

region          = "me-jeddah-1"
config_file_profile = "DEFAULT"
compartment_ocid = "ocid1.compartment.oc1..aaaaaaaaXXXXXXXXXX"
enable_oke       = false # Part 1; flipped to true for Part 2
vcn_cidr         = "10.2.0.0/16"
oke_cni_type      = "OCI_VCN_IP_NATIVE"
oke_node_pool_size = 1
k8s_namespace    = "w3-demo"

```

## 6. Terraform Design Standards Applied

The stack follows a consistent set of module-design conventions carried through the Cloud Build track: changeable values are isolated from structure, resources are grouped by relationship rather than wrapped one-to-one, and the application layer stays outside of Terraform state.

| Principle                                       | Implementation in this stack                                                                                                                                                                                             |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variables-driven configuration                  | Changeable arguments in variables.tf; environment values in tfvars; only the example file is committed                                                                                                                   |
| Related-group modules                           | network (VCN + Internet Gateway + NAT Gateway + Service Gateway + shared log group); subnet (subnet + route table + security list + logs) × 3; oke (cluster + native CNI + managed pool + NSGs in nsg.tf) — see Figure 3 |
| No thin one-resource wrappers                   | Gateways live in modules/network; NSGs live in modules/oke; there is no one-resource wrapper per gateway or NSG                                                                                                          |
| Multi-file root layout                          | versions.tf → providers.tf → variables.tf → locals.tf → network.tf → subnets.tf → oke.tf → outputs.tf                                                                                                                    |
| Application layer separated from infrastructure | k8s/ is applied with kubectl only after the cluster reaches ACTIVE                                                                                                                                                       |
| Documentation as a finished deliverable         | This document is the single Terraform and evidence narrative for Week 3, covering code, commands, evidence, diagrams, and the destroy                                                                                    |

## 7. Commands Executed (Successful Path)

Working directory: terraform/nonprd/week3-containerized-oke. Profile DEFAULT, region me-jeddah-1, Windows PowerShell.

### 7.1 Tooling baseline

```
terraform version
→ Terraform v1.15.8 + provider registry.terraform.io/oracle/oci v6.37.0
```

### 7.2 Part 1 — network (enable\_oke = false)

*Captured 2026-08-15, approximately 18:53 +03.*

```
terraform init
terraform fmt -recursive
terraform validate
terraform plan
terraform apply
# Apply complete! Resources: 16 added, 0 changed, 0 destroyed.
# No oci_containerengine_*
```

The 16 managed resources were: the VCN, Internet Gateway, NAT Gateway, Service Gateway, log group, three subnet + route table + security list groups, and the lb subnet's capture filter and flow log.

### 7.3 Part 2 — OKE (enable\_oke = true)

*Captured 2026-08-16, approximately 00:24 +03, after the intern's create permission on the cluster-family / OKE service policy was granted (Section 10.1).*

```
# terraform.tfvars: enable_oke = true, ssh_public_key set
terraform plan
terraform apply
# Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
# cluster w3-oke (~8m40s)
# node pool w3-np (~5m26s)
```

Cluster OCID: ocid1.cluster.oc1.me-jeddah-1.aaaaaaaavhl7onzm5htuw5k2reau64kdc7effu4itsy5qyztsclhoyslgma

Node pool OCID: ocid1.nodepool.oc1.me-jeddah-1.aaaaaaaamuhzibv3uogyvmnn753xsc5e4bbkt3s6furajrrelnmehg75g3sa

The OKE NSGs were already present in the module's state; this apply added only the cluster and the node pool.

### 7.4 Workload deployment via kubectl

*The request path exercised here is the one traced in Figure 2 (Section 3.2).*

```
$env:KUBECONFIG = "$env:USERPROFILE\.kube\w3-oke"

oci ce cluster create-kubeconfig `
  --cluster-id ocid1.cluster.oc1.me-jeddah-1.aaaaaaaavhl7onzm5htuw5k2reau64kdc7effu4itsy5qyztsclhoyslgma `
  --file $env:USERPROFILE\.kube\w3-oke `
  --region me-jeddah-1 `
  --token-version 2.0.0 `
  --kube-endpoint PUBLIC_ENDPOINT
```

```
kubectl apply -f k8s/namespace.yaml
kubectl apply -f k8s/pvc.yaml
kubectl apply -f k8s/deployment.yaml
kubectl apply -f k8s/service.yaml
```

```
kubectl -n w3-demo get pods,deploy
→ pod/nginx-... 1/1 Running
→ deployment.apps/nginx 1/1
```

```
kubectl -n w3-demo get pvc nginx-data
→ Bound 50Gi RWX oci-bv volume csi-082c7776-ed2e-4301-80e1-fd4d7cf02fe2
```

```
kubectl -n w3-demo get svc nginx
→ LoadBalancer EXTERNAL-IP 193.122.68.107 80:31785/TCP
```

```
kubectl -n w3-demo exec deploy/nginx -c nginx -- df -h /data
→ /dev/sdb 48.9G mounted on /data
```

Browser check: <http://193.122.68.107/> returned the nginx welcome page.

## 7.5 Destroy

*Captured 2026-08-16, approximately 01:18–01:26 +03. Kubernetes objects sit outside Terraform state, so they were removed first.*

```
kubectl -n w3-demo delete svc nginx
kubectl -n w3-demo delete deploy nginx
kubectl -n w3-demo delete pvc nginx-data
kubectl delete ns w3-demo
terraform destroy -auto-approve -no-color
# Destroy complete! Resources: 36 destroyed.
# terraform state list → empty
```

## 8. Evidence Catalogue

The raw evidence files back the excerpts already quoted through this document; the substance is summarized below.

| File                               | What it proves                                                                       |
|------------------------------------|--------------------------------------------------------------------------------------|
| terraform-part1-network-apply.txt  | Part 1 apply added 16 resources; enable_oke = false; VCN 10.2.0.0/16                 |
| terraform-part1-outputs.txt        | Part 1 outputs (subnet IDs, log group, Service Gateway); OKE outputs null            |
| terraform-part1-state-list.txt     | 16 managed addresses; no oci_containerengine_*                                       |
| oci-cli-part1-network-snapshot.txt | Live w3-* VCN, gateways, subnets, route tables, security lists, log group            |
| compartment-intern-18-audit.txt    | All oci_* resources use the intern-18 compartment OCID                               |
| terraform-part2-oke-apply.txt      | Cluster and node pool added (2 resources); CNI OCI_VCN_IP_NATIVE; Kubernetes v1.34.1 |
| terraform-part2-oke-outputs.txt    | Cluster/node-pool IDs, NSG IDs, kubeconfig hint                                      |
| terraform-part2-oke-state-list.txt | Cluster, node pool, NSGs, three subnet modules                                       |
| oci-cli-part2-oke-snapshot.txt     | w3-oke ACTIVE, w3-np ACTIVE size 1, three NSGs                                       |
| terraform-part2-oke-destroy.txt    | 36 resources destroyed, exit 0, empty state; kubectl cleanup ran first               |
| oci-cli-post-destroy-inventory.txt | intern-18 shows zero live billable resources; TERMINATED / DELETED ghosts only       |

### 8.1 Key excerpts

#### Part 1 state (abbreviated):

```
oci_core_vcn.this
oci_core_internet_gateway.this
oci_core_nat_gateway.this
oci_core_service_gateway.this[0]
oci_logging_log_group.vcn_flow[0]
module.subnet["lb"].oci_core_subnet.this
module.subnet["lb"].oci_logging_log.flow[0]
module.subnet["workers"].oci_core_subnet.this
module.subnet["pods"].oci_core_subnet.this
# no oci_containerengine_*
```

#### Part 2 OCI CLI (highlights):

```
w3-oke  ACTIVE  BASIC_CLUSTER  v1.34.1
        ocid1.cluster.oc1.me-jeddah-
1.aaaaaaaavhl7onzm5htuw5k2reau64kdc7effu4itsy5qyztsclhoyslgma
w3-np   ACTIVE  VM.Standard.E4.Flex  size=1
NSGs    w3-nsg-oke-api / w3-nsg-oke-workers / w3-nsg-oke-pods  AVAILABLE
```

#### Part 2 outputs (highlights):

```
oke_cluster_id      = ocid1.cluster.oc1.me-jeddah-
1.aaaaaaaavhl7onzm5htuw5k2reau64kdc7effu4itsy5qyztsclhoyslgma
oke_cni_type        = OCI_VCN_IP_NATIVE
kubernetes_version  = v1.34.1
oke_node_pool_id    = ocid1.nodepool.oc1.me-jeddah-
1.aaaaaaaamuhzibv3uogyvmnn753xsc5e4bbkt3s6furajrrelnmehg75g3sa
```

| Stage              | Resource count                                 |
|--------------------|------------------------------------------------|
| Part 1 apply added | 16                                             |
| Part 2 apply added | 2 (cluster + node pool; NSGs already in state) |
| Destroy            | 36 destroyed                                   |

Workload validation: pods Running 1/1; PVC Bound 50Gi on oci-bv; the browser reached <http://193.122.68.107/> and returned the nginx welcome page; df -h /data inside the pod showed /dev/sdb mounted with roughly 48.9G available.

## 9. Screenshot Evidence

The screenshots below were captured from the OCI Console and the kubectl CLI after each Terraform stage. Session tokens and kubeconfig tokens are cropped out of every image; no secrets, PEM paths, or terraform.tfvars content appear in any capture. A small number of inventory-list screenshots (a combined subnet list, a combined route-table list, and the VCN IP-administration view) were captured alongside these but are not repeated here, since every resource they show is already covered individually below. The node-pool worker is likewise covered by the node-pools tab (B02b) and the worker instance detail (B04) rather than a separate Nodes-tab capture.

### 9.1 Stage A — network

#### A00 — Resource Explorer inventory

| Resource type          | Region name                | OCID         | Compartment                   | Status    | Time created               |
|------------------------|----------------------------|--------------|-------------------------------|-----------|----------------------------|
| Logs                   | Saudi Arabia West (Jeddah) | ...gpgnduura | intern-18-salah-abdelhady-cmp | Active    | Aug 15, 2026, 15:53:58 UTC |
| Subnets                | Saudi Arabia West (Jeddah) | ...c55uuvra  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:55 UTC |
| Subnets                | Saudi Arabia West (Jeddah) | ...v835x2ga  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:54 UTC |
| Subnets                | Saudi Arabia West (Jeddah) | ...d3pfhnd4q | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:54 UTC |
| Route Tables           | Saudi Arabia West (Jeddah) | ...h127uq3a  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:53 UTC |
| Security Lists         | Saudi Arabia West (Jeddah) | ...3m38mumpu | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:53 UTC |
| Route Tables           | Saudi Arabia West (Jeddah) | ...c3bvkvdka | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:53 UTC |
| Security Lists         | Saudi Arabia West (Jeddah) | ...vg75fmgq  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:53 UTC |
| Security Lists         | Saudi Arabia West (Jeddah) | ...e44kmhmq  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:53 UTC |
| Route Tables           | Saudi Arabia West (Jeddah) | ...p02gvr5q  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:53 UTC |
| Public IPs             | Saudi Arabia West (Jeddah) | ...p0mty4pqa | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:49 UTC |
| Service Gateways       | Saudi Arabia West (Jeddah) | ...m0p0c3u7a | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:49 UTC |
| NAT Gateways           | Saudi Arabia West (Jeddah) | ...3baq5bba  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:48 UTC |
| Internet Gateways      | Saudi Arabia West (Jeddah) | ...gg23mrv4q | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:46 UTC |
| Route Tables           | Saudi Arabia West (Jeddah) | ...2ap87mbda | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:41 UTC |
| Security Lists         | Saudi Arabia West (Jeddah) | ...3vq37q3qa | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:41 UTC |
| Virtual Cloud Networks | Saudi Arabia West (Jeddah) | ...m0p0c3u7a | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026, 15:53:41 UTC |
| Log Groups             | Saudi Arabia West (Jeddah) | ...v835x2ga  | intern-18-salah-abdelhady-cmp | Active    | Aug 15, 2026, 15:53:41 UTC |

**A00** — Resource Explorer showing every w3-\* resource as Available or Active.

#### A01 — VCN w3-vcn

| VCN Information              |                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Compartment                  | intern-18-salah-abdelhady-cmp                                                                                     |
| Created                      | Aug 15, 2026, 15:53:41 UTC                                                                                        |
| IPv4 CIDR Block              | 10.2.0.0/16                                                                                                       |
| IPv6 Prefix                  | —                                                                                                                 |
| Oracle-allocated IPv6 Prefix | —                                                                                                                 |
| BYOIP IPv6 Prefixes          | —                                                                                                                 |
| ULA IPv6 Prefixes            | —                                                                                                                 |
| OCID                         | ocid1.vcn.oc1.me-jeddah-1.amaaaaaavjakbniajs4reyldflkic74y4rbuflavqdgcoimcepgs4ca <a href="#">Copy</a>            |
| DNS Resolver                 | ocid1.dnsresolver.oc1.me-jeddah-1.amaaaaaavjakbniajrqllp3j3vnfxi7iblnr4t4nntqsubcfxirtg5x34q <a href="#">Copy</a> |
| Default Route Table          | Default Route Table for w3-vcn                                                                                    |
| DNS Domain Name              | w3vcn.oraclevcn.com                                                                                               |

**A01** — VCN w3-vcn Available, CIDR 10.2.0.0/16.



## A02 — gateways

**Internet Gateways**

Search and Filter

Applied filters: `Compartment: intern-18-salah-abdelhady-cmp`

Create Internet Gateway

| Name   | State     | Route Table | Created                    |
|--------|-----------|-------------|----------------------------|
| w3-igw | Available | —           | Aug 15, 2026, 15:53:46 UTC |

Page 1 of 1 (1 - 1 of 1 total items)

Items per page: 25

**A02 — Internet Gateway w3-igw Available.**

**NAT Gateways**

Search and Filter

Applied filters: `Compartment: intern-18-salah-abdelhady-cmp`

Create NAT Gateway

| Name   | State     | Public IP address | Route Table | Created                    |
|--------|-----------|-------------------|-------------|----------------------------|
| w3-nat | Available | 130.110.115.23    | —           | Aug 15, 2026, 15:53:48 UTC |

Page 1 of 1 (1 - 1 of 1 total items)

Items per page: 25

**A02 — NAT Gateway w3-nat Available.**

**Service Gateways**

Search and Filter

Applied filters: `Compartment: intern-18-salah-abdelhady-cmp`

Create Service Gateway

| Name   | State     | Services                                    | Route Table | Created                    |
|--------|-----------|---------------------------------------------|-------------|----------------------------|
| w3-sgw | Available | All JED Services In Oracle Services Network | —           | Aug 15, 2026, 15:53:49 UTC |

Page 1 of 1 (1 - 1 of 1 total items)

Items per page: 25

**A02 — Service Gateway w3-sgw Available, routed to All JED Services In Oracle Services Network.**

## A03–A05 — subnets

Cloud

Search resources, services, documentation, and Marketplace

**Networking**

Overview

Virtual cloud networks

Web Application Acceleration

Load balancers

DNS management

Customer connectivity

Cluster Placement Groups

IP management

Network Command Center

← w3-vcn

**w3-lb-subnet** Available

Subnet

Details IP administration Private service access Security Monitoring Tags

**Subnet Information**

|                              |                                                                                        |      |
|------------------------------|----------------------------------------------------------------------------------------|------|
| OCID                         | ... 1.subnet.oc1.me-jeddah-1.aaaaaaa2zvm2pgt3f7j5ggeguswxcvzden155rgd4bnhdhsdqzhieub4q | Copy |
| IPv4 CIDR Block              | 10.2.1.0/24                                                                            |      |
| IPv4 CIDR Blocks             | 10.2.1.0/24                                                                            |      |
| Oracle-allocated IPv6 Prefix | —                                                                                      |      |
| BYOIP IPv6 Prefixes          | —                                                                                      |      |
| ULA IPv6 Prefixes            | —                                                                                      |      |
| Virtual Router MAC Address   | 00:00:17:71:36:06                                                                      |      |
| Subnet Type                  | Regional                                                                               |      |
| Availability Domain          | —                                                                                      |      |
| Compartment                  | intern-18-salah-abdelhady-cmp                                                          |      |
| DNS Domain Name              | w3lb.w3vcn.oraclevcn.com                                                               | Copy |

**A03 — w3-lb-subnet, public, 10.2.1.0/24.**

The screenshot shows the Oracle Cloud console interface. On the left is a navigation menu with 'Networking' selected. The main area displays the 'w3-workers-subnet' details. The 'Subnet Information' section is expanded, showing various attributes.

| Attribute                    | Value                                                                                 | Action |
|------------------------------|---------------------------------------------------------------------------------------|--------|
| OCID                         | ocid1.subnet.oc1.me-jeddah-1.aaaaaaaqkheqctvg4rg24gtejkd5v3bxrhdw5zxo6ykd4h5odt3kt2qa | Copy   |
| IPv4 CIDR Block              | 10.2.10.0/24                                                                          |        |
| IPv4 CIDR Blocks             | 10.2.10.0/24                                                                          |        |
| Oracle-allocated IPv6 Prefix | —                                                                                     |        |
| BYOIP IPv6 Prefixes          | —                                                                                     |        |
| ULA IPv6 Prefixes            | —                                                                                     |        |
| Virtual Router MAC Address   | 00:00:17:71:36:06                                                                     |        |
| Subnet Type                  | Regional                                                                              |        |
| Availability Domain          | —                                                                                     |        |
| Compartment                  | intern-18-salah-abdelhady-cmp                                                         |        |
| DNS Domain Name              | w3workers.w3vcn.oraclevn.com                                                          | Copy   |

**A04** — *w3-workers-subnet*, private, 10.2.10.0/24.

The screenshot shows the Oracle Cloud console interface. On the left is a navigation menu with 'Networking' selected. The main area displays the 'w3-pods-subnet' details. The 'Subnet Information' section is expanded, showing various attributes.

| Attribute                    | Value                                                                                    | Action |
|------------------------------|------------------------------------------------------------------------------------------|--------|
| OCID                         | ocid1.subnet.oc1.me-jeddah-1.aaaaaaaakq4u274mtq6o6hvo6qla4ptabc4s2l74haxnm13yggcdzuxvrva | Copy   |
| IPv4 CIDR Block              | 10.2.128.0/18                                                                            |        |
| IPv4 CIDR Blocks             | 10.2.128.0/18                                                                            |        |
| Oracle-allocated IPv6 Prefix | —                                                                                        |        |
| BYOIP IPv6 Prefixes          | —                                                                                        |        |
| ULA IPv6 Prefixes            | —                                                                                        |        |
| Virtual Router MAC Address   | 00:00:17:71:36:06                                                                        |        |
| Subnet Type                  | Regional                                                                                 |        |
| Availability Domain          | —                                                                                        |        |
| Compartment                  | intern-18-salah-abdelhady-cmp                                                            |        |
| DNS Domain Name              | w3pods.w3vcn.oraclevn.com                                                                | Copy   |

**A05** — *w3-pods-subnet*, private, 10.2.128.0/18.

## A06–A08 — route tables

**w3-lb-subnet-rt** Available  
Route Table

Details **Route Rules** Tags

**Route Rules**

Traffic within the VCN is handled by the VCN's local routing by default. Intra-VCN routing allows you more control over routing between subnets. [Learn more](#). If you're having problems, use [Network Path Analyzer](#) to check your connections.

Search and Filter

Add Route Rules Actions

| <input type="checkbox"/> | Destination ^ | Target Type      | Target | Route Type | Description                       |
|--------------------------|---------------|------------------|--------|------------|-----------------------------------|
| <input type="checkbox"/> | 0.0.0.0/0     | Internet Gateway | w3-igw | Static     | Default route to Internet Gateway |

Page 1 of 1 (1 - 1 of 1 total items)

A06 — w3-lb-subnet-rt — default route 0.0.0.0/0 to Internet Gateway w3-igw.

**w3-workers-subnet-rt** Available  
Route Table

Details **Route Rules** Tags

**Route Rules**

Traffic within the VCN is handled by the VCN's local routing by default. Intra-VCN routing allows you more control over routing between subnets. [Learn more](#). If you're having problems, use [Network Path Analyzer](#) to check your connections.

Search and Filter

Add Route Rules Actions

| <input type="checkbox"/> | Destination ^                               | Target Type     | Target | Route Type | Description                                     |
|--------------------------|---------------------------------------------|-----------------|--------|------------|-------------------------------------------------|
| <input type="checkbox"/> | 0.0.0.0/0                                   | NAT Gateway     | w3-nat | Static     | Default route to NAT Gateway                    |
| <input type="checkbox"/> | All JED Services in Oracle Services Network | Service Gateway | w3-sgw | Static     | All Oracle Services Network via Service Gateway |

Page 1 of 1 (1 - 2 of 2 total items)

A07 — w3-workers-subnet-rt — routes through NAT w3-nat and Service Gateway w3-sgw.

**w3-pods-subnet-rt** Available  
Route Table

Details **Route Rules** Tags

**Route Rules**

Traffic within the VCN is handled by the VCN's local routing by default. Intra-VCN routing allows you more control over routing between subnets. [Learn more](#). If you're having problems, use [Network Path Analyzer](#) to check your connections.

Search and Filter

Add Route Rules Actions

| <input type="checkbox"/> | Destination ^                               | Target Type     | Target | Route Type | Description                                     |
|--------------------------|---------------------------------------------|-----------------|--------|------------|-------------------------------------------------|
| <input type="checkbox"/> | 0.0.0.0/0                                   | NAT Gateway     | w3-nat | Static     | Default route to NAT Gateway                    |
| <input type="checkbox"/> | All JED Services in Oracle Services Network | Service Gateway | w3-sgw | Static     | All Oracle Services Network via Service Gateway |

Page 1 of 1 (1 - 2 of 2 total items)

A08 — w3-pods-subnet-rt — routes through NAT w3-nat and Service Gateway w3-sgw.

## A09–A11 — security lists

**w3-lb-subnet-sl** Available

Security List

Instance traffic is controlled by firewall rules on each instance in addition to this Security List

Details Security rules Tags

### Ingress Rules

Q Search and Filter

Add Ingress Rules Actions

| <input type="checkbox"/> | Statusless | Source             | IP Protocol | Source Port Range | Destination Port Range | Type and Code | Allows                                                                                          | Description                      |
|--------------------------|------------|--------------------|-------------|-------------------|------------------------|---------------|-------------------------------------------------------------------------------------------------|----------------------------------|
| <input type="checkbox"/> | No         | 0.0.0.0/0          | TCP         | All               | 443                    |               | TCP traffic for ports: 443 HTTPS                                                                | HTTPS from internet              |
| <input type="checkbox"/> | No         | 10.2.0.0/16        | ICMP        |                   |                        | 3             | ICMP traffic for: 3 Destination Unreachable                                                     | ICMP type 3 within VCN           |
| <input type="checkbox"/> | No         | 197.164.1.14.44/32 | TCP         | All               | 6443                   |               | TCP traffic for ports: 6443                                                                     | Kubernetes API from laptop       |
| <input type="checkbox"/> | No         | 10.2.128.0/18      | TCP         | All               | 6443                   |               | TCP traffic for ports: 6443                                                                     | Kubernetes API from pods         |
| <input type="checkbox"/> | No         | 0.0.0.0/0          | TCP         | All               | 80                     |               | TCP traffic for ports: 80                                                                       | HTTP from internet to Service LB |
| <input type="checkbox"/> | No         | 10.2.10.0/24       | TCP         | All               | 12250                  |               | TCP traffic for ports: 12250                                                                    | OSX control from workers         |
| <input type="checkbox"/> | No         | 0.0.0.0/0          | ICMP        |                   |                        | 3, 4          | ICMP traffic for: 3, 4 Destination Unreachable: Fragmentation Needed and Don't Fragment was Set | ICMP type 3 code 4 (PMTU)        |
| <input type="checkbox"/> | No         | 10.2.10.0/24       | TCP         | All               | 6443                   |               | TCP traffic for ports: 6443                                                                     | Kubernetes API from workers      |

Page 1 of 1 (1 - 8 of 8 total items)

Items per page: 25

### Egress Rules

Q Search and Filter

Add Egress Rules Actions

| <input type="checkbox"/> | Statusless | Destination | IP Protocol   | Source Port Range | Destination Port Range | Type and Code | Allows                    | Description      |
|--------------------------|------------|-------------|---------------|-------------------|------------------------|---------------|---------------------------|------------------|
| <input type="checkbox"/> | No         | 0.0.0.0/0   | All Protocols |                   |                        |               | All traffic for all ports | Allow all egress |

**A09** — *w3-lb-subnet-sl* — HTTP/HTTPS plus the Kubernetes API ports 6443 and 12250.

**w3-workers-subnet-sl** Available

Security List

Instance traffic is controlled by firewall rules on each instance in addition to this Security List

Details Security rules Tags

### Ingress Rules

Q Search and Filter

Add Ingress Rules Actions

| <input type="checkbox"/> | Statusless | Source             | IP Protocol   | Source Port Range | Destination Port Range | Type and Code | Allows                                                                                          | Description                |
|--------------------------|------------|--------------------|---------------|-------------------|------------------------|---------------|-------------------------------------------------------------------------------------------------|----------------------------|
| <input type="checkbox"/> | No         | 10.2.0.0/16        | ICMP          |                   |                        | 3             | ICMP traffic for: 3 Destination Unreachable                                                     | ICMP type 3 within VCN     |
| <input type="checkbox"/> | No         | 10.2.10.0/24       | TCP           | All               | 10250                  |               | TCP traffic for ports: 10250                                                                    | Kubelet from API/LB subnet |
| <input type="checkbox"/> | No         | 10.2.10.0/24       | TCP           | All               | 80                     |               | TCP traffic for ports: 80                                                                       | Service LB to app port     |
| <input type="checkbox"/> | No         | 197.164.1.14.44/32 | TCP           | All               | 22                     |               | TCP traffic for ports: 22 SSH Remote Login Protocol                                             | SSH from allowed_ssh_cidr  |
| <input type="checkbox"/> | No         | 10.2.10.0/24       | TCP           | All               | 30000-32767            |               | TCP traffic for ports: 30000-32767                                                              | Service LB NodePort range  |
| <input type="checkbox"/> | No         | 10.2.10.0/24       | All Protocols |                   |                        |               | All traffic for all ports                                                                       | All from worker subnet     |
| <input type="checkbox"/> | No         | 10.2.128.0/18      | All Protocols |                   |                        |               | All traffic for all ports                                                                       | All from pod subnet        |
| <input type="checkbox"/> | No         | 0.0.0.0/0          | ICMP          |                   |                        | 3, 4          | ICMP traffic for: 3, 4 Destination Unreachable: Fragmentation Needed and Don't Fragment was Set | ICMP type 3 code 4 (PMTU)  |

Page 1 of 1 (1 - 8 of 8 total items)

Items per page: 25

### Egress Rules

Q Search and Filter

Add Egress Rules Actions

| <input type="checkbox"/> | Statusless | Destination | IP Protocol   | Source Port Range | Destination Port Range | Type and Code | Allows                    | Description      |
|--------------------------|------------|-------------|---------------|-------------------|------------------------|---------------|---------------------------|------------------|
| <input type="checkbox"/> | No         | 0.0.0.0/0   | All Protocols |                   |                        |               | All traffic for all ports | Allow all egress |

**A10** — *w3-workers-subnet-sl* — kubelet 10250, application port 80, NodePort range, and SSH.

**w3-pods-subnet-sl** Available

Security List

Instance traffic is controlled by firewall rules on each instance in addition to this Security List

Details Security rules Tags

### Ingress Rules

Q Search and Filter

Add Ingress Rules Actions

| <input type="checkbox"/> | Statusless | Source        | IP Protocol   | Source Port Range | Destination Port Range | Type and Code | Allows                                                                                          | Description               |
|--------------------------|------------|---------------|---------------|-------------------|------------------------|---------------|-------------------------------------------------------------------------------------------------|---------------------------|
| <input type="checkbox"/> | No         | 10.2.0.0/16   | ICMP          |                   |                        | 3             | ICMP traffic for: 3 Destination Unreachable                                                     | ICMP type 3 within VCN    |
| <input type="checkbox"/> | No         | 10.2.10.0/24  | All Protocols |                   |                        |               | All traffic for all ports                                                                       | All from worker subnet    |
| <input type="checkbox"/> | No         | 10.2.128.0/18 | All Protocols |                   |                        |               | All traffic for all ports                                                                       | All from pod subnet       |
| <input type="checkbox"/> | No         | 0.0.0.0/0     | ICMP          |                   |                        | 3, 4          | ICMP traffic for: 3, 4 Destination Unreachable: Fragmentation Needed and Don't Fragment was Set | ICMP type 3 code 4 (PMTU) |

Page 1 of 1 (1 - 4 of 4 total items)

Items per page: 25

### Egress Rules

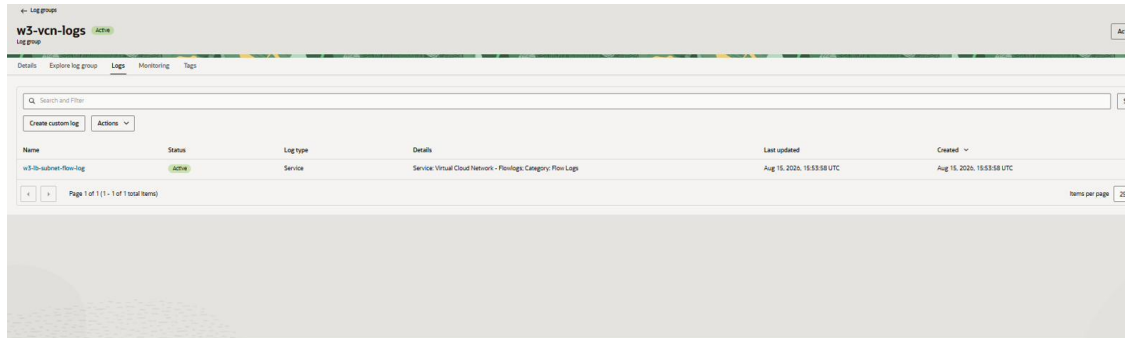
Q Search and Filter

Add Egress Rules Actions

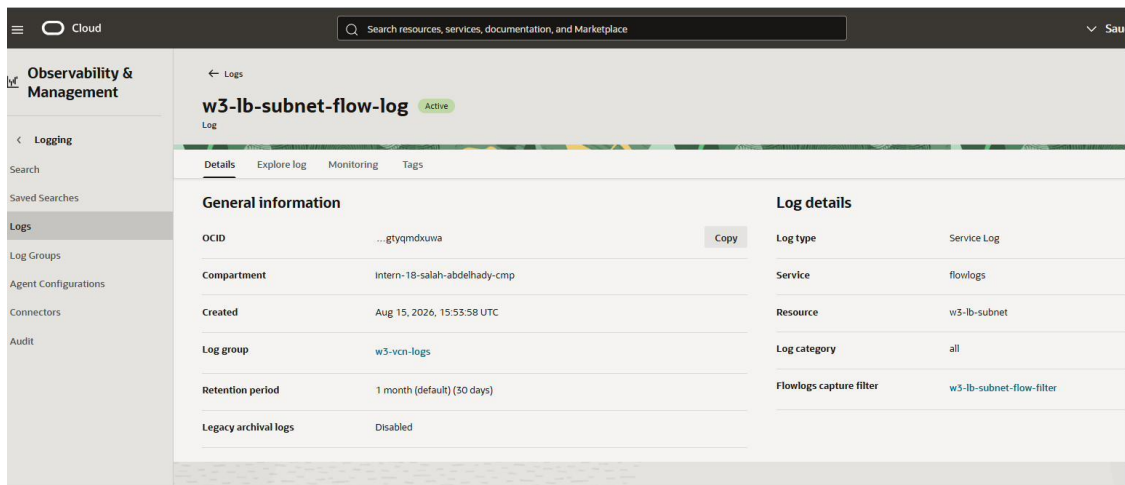
| <input type="checkbox"/> | Statusless | Destination | IP Protocol   | Source Port Range | Destination Port Range | Type and Code | Allows                    | Description      |
|--------------------------|------------|-------------|---------------|-------------------|------------------------|---------------|---------------------------|------------------|
| <input type="checkbox"/> | No         | 0.0.0.0/0   | All Protocols |                   |                        |               | All traffic for all ports | Allow all egress |

**A11** — *w3-pods-subnet-sl* — all traffic from the workers and pods subnets.

## A12–A13 — logging

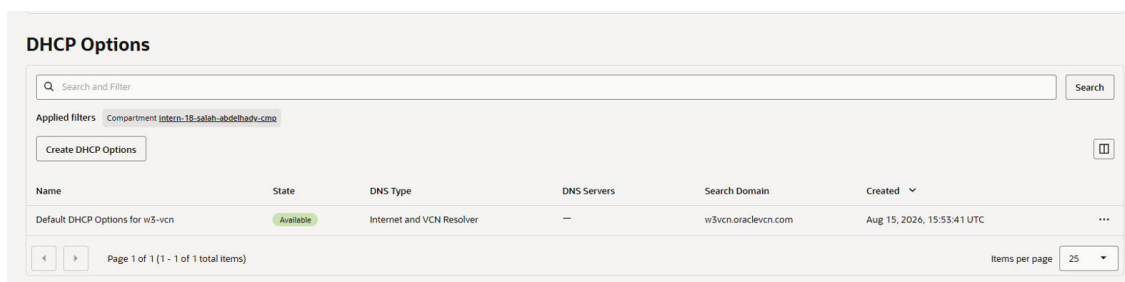


**A12** — Log group *w3-vcn-logs*, Active.



**A13** — Flow log *w3-lb-subnet-flow-log*, Active on *w3-lb-subnet*.

## A99 — DHCP options



**A99** — Default DHCP options for *w3-vcn*.

## 9.2 Stage B — OKE

### B01 — cluster detail

**Cluster details**

|                               |                                                         |
|-------------------------------|---------------------------------------------------------|
| Cluster status                | Active                                                  |
| Node pools                    | 1                                                       |
| Cluster ID                    | ...aavh7onzm5htuw5k2reau64kdc7effu4itsy5qyzscilhoyslgma |
| Compartment                   | intern-18-salah-abdelhady-cmp                           |
| Launched                      | Aug 15, 2026, 21:08:50 UTC                              |
| Created                       | -                                                       |
| Kubernetes version            | v1.34.1 <span>Upgrade recommended</span>                |
| Secrets encryption key        | Oracle-managed                                          |
| Cluster type                  | Basic                                                   |
| Cluster credentials expire on | 8/15/2031                                               |

**Network information**

|                                      |                                                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------------------------------|
| VCN name                             | w3-vcn                                                                                                |
| VCN ID                               | ...-1.amaaaaaavjakbnia5adnxhzs6uqk56l2zrvckckfsuzlpzn3lbmio7rq5a                                      |
| Compartment                          | intern-18-salah-abdelhady-cmp                                                                         |
| Services CIDR                        | 10.96.0.0/16                                                                                          |
| Network type                         | OCI_VCN_IP_NATIVE                                                                                     |
| Kubernetes API endpoint subnet       | w3-lb-subnet                                                                                          |
| Kubernetes API private IPv4 endpoint | 10.2.1.203:6443                                                                                       |
| Kubernetes API IPv6 endpoint         | Not assigned                                                                                          |
| Kubernetes API public endpoint       | 80.225.71.220:6443                                                                                    |
| Network security group               | ocid1.networksecuritygroup.oc1.me-jeddah-1.aaaaaaaamuhzibv5uogvymnn753xsc5e4bkt356furajrelnmehg75g5sa |

**B01** — Cluster *w3-oke* Active, network type *OCI\_VCN\_IP\_NATIVE*, Kubernetes *v1.34.1*.

### B02 — node pool

**Node pool details**

|                         |                                                              |
|-------------------------|--------------------------------------------------------------|
| Node pool status        | Active                                                       |
| Node pool ID            | ...aaaaaaamuhzibv5uogvymnn753xsc5e4bkt356furajrelnmehg75g5sa |
| Node type               | Managed                                                      |
| Compartment             | intern-18-salah-abdelhady-cmp                                |
| Network security groups | w3-nsg-oke-workers                                           |
| Custom boot volume size | 50                                                           |
| OCPUs                   | 1                                                            |
| Amount of memory (GB)   | 8                                                            |
| Kubernetes version      | v1.34.1 <span>Upgrade recommended</span>                     |
| Shape                   | VM.Standard.E4.Flex                                          |
| Image name              | Oracle-Linux-8.10-2025.09.16-0-OKE-1.34.1-1330               |

**Placement configurations**

|                    |                      |
|--------------------|----------------------|
| AD 1               | oXVtME-JEDDAH-1-AD-1 |
| Worker node subnet | w3-workers-subnet    |
| Capacity type      | On-demand capacity   |

**Pod communication**

|                         |                 |
|-------------------------|-----------------|
| Subnet                  | w3-pods-subnet  |
| Network security groups | w3-nsg-oke-pods |
| Number of pods per node | 31              |

**Node cycling information**

|              |               |
|--------------|---------------|
| Cycle option | Replace nodes |
|--------------|---------------|

**B02** — Node pool *w3-np* Active, shape *VM.Standard.E4.Flex* (1 OCPU / 8 GB), with the worker and pod subnets and NSGs attached.

## B02b — node pools tab

**Node pools**

Accessing your private nodes  
In order to access your private nodes with a public SSH key you will need to set up a bastion host (a.k.a. jump box). [Learn more about setting up a bastion host](#)

Search and Filter

Applied filters: Compartment: intern-18-salah-abdelhady-cmp

Add node pool

| Node pool | Node pool status | Kubernetes version | Kubernetes status   | Image name                                     | Total worker nodes |
|-----------|------------------|--------------------|---------------------|------------------------------------------------|--------------------|
| w3-np     | Active           | v1.34.1            | Upgrade recommended | Oracle-Linux-8.10-2025.09.16-0-OKE-1.34.1-1330 | 1                  |

Page 1 of 1 (1 - 1 of 1 total items)

Items per page: 10

**B02b** — Cluster Node pools tab: w3-np Active, one worker.

## B04 — worker instance

**General information**

|                     |                                                             |
|---------------------|-------------------------------------------------------------|
| Availability domain | AD-1                                                        |
| Fault domain        | FD-1                                                        |
| Region              | me-jeddah-1                                                 |
| OCID                | ...h-1amgklyrjakbncdc42cpdasimmgqnuik3uckew5ndfmgdbijayokfq |
| Launched            | Aug 15, 2026, 21:17:42 UTC                                  |
| Compartment         | intern-18-salah-abdelhady-cmp                               |
| Capacity type       | On-demand                                                   |

**Launch options**

|                         |                 |
|-------------------------|-----------------|
| NIC attachment type     | PARAVIRTUALIZED |
| Remote data volume      | PARAVIRTUALIZED |
| Firmware                | UEFI_64         |
| Boot volume type        | PARAVIRTUALIZED |
| In-transit encryption   | Disabled        |
| Secure Boot             | Disabled        |
| Measured Boot           | Disabled        |
| Trusted Platform Module | Disabled        |

**Shape configuration**

|       |                     |
|-------|---------------------|
| Shape | VM.Standard.E4.Flex |
|-------|---------------------|

**B04** — Worker instance Running in the private subnet only, AD-1 in me-jeddah-1, private IP 10.2.10.193.

## B05 — OKE NSGs

**Resource Explorer**

Search for resources in this tenancy

Search and Filter

Applied filters: Compartment: All, Region name: Saudi Arabia West (Jeddah)

Send as resource collection

Actions

| Display name          | Resource type           | Region name                | OCID           | Compartment                   | Status    | Time created |
|-----------------------|-------------------------|----------------------------|----------------|-------------------------------|-----------|--------------|
| w3-nsg-oke-pods       | Network Security Groups | Saudi Arabia West (Jeddah) | ...perkerru7mq | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026 |
| w3-nsg-oke-workers    | Network Security Groups | Saudi Arabia West (Jeddah) | ...logklylra   | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026 |
| w3-nsg-oke-api        | Network Security Groups | Saudi Arabia West (Jeddah) | ...wmdstf52ka  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026 |
| w3-lb-subnet-flow-log | Logs                    | Saudi Arabia West (Jeddah) | ...gtyrmdouma  | intern-18-salah-abdelhady-cmp | Active    | Aug 15, 2026 |
| w3-pods-subnet        | Subnets                 | Saudi Arabia West (Jeddah) | ...cdzuvvna    | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026 |
| w3-workers-subnet     | Subnets                 | Saudi Arabia West (Jeddah) | ...odt3kt2qa   | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026 |
| w3-lb-subnet          | Subnets                 | Saudi Arabia West (Jeddah) | ...dgzhieub4q  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026 |
| w3-pods-subnet-rt     | Route Tables            | Saudi Arabia West (Jeddah) | ...n62foajisa  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026 |
| w3-lb-subnet-sl       | Security Lists          | Saudi Arabia West (Jeddah) | ...3m3fdmumpa  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026 |
| w3-lb-subnet-rt       | Route Tables            | Saudi Arabia West (Jeddah) | ...cjxbkwvdkx  | intern-18-salah-abdelhady-cmp | Available | Aug 15, 2026 |

**B05** — NSGs w3-nsg-oke-api, w3-nsg-oke-workers, and w3-nsg-oke-pods, all Available.



## 9.3 Stage C — workload and destroy

### C01 — pods and deployment

```
PS E:\work\Ejada\terraform\nonprd\week3-containerized-oke> $env:KUBECONFIG = "$env:USERPROFILE\.kube\w3-oke"
PS E:\work\Ejada\terraform\nonprd\week3-containerized-oke> kubectl -n w3-demo get pods,deploy
NAME                                READY   STATUS    RESTARTS   AGE
pod/nginx-58749cf6cc-hp8rz         1/1     Running   0           7m20s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx              1/1     1             1           10m
PS E:\work\Ejada\terraform\nonprd\week3-containerized-oke>
```

**C01** — `kubectl -n w3-demo get pods,deploy` — pod Running 1/1, deployment Available.

### C02 — PVC bound

```
PS E:\work\Ejada\terraform\nonprd\week3-containerized-oke> kubectl -n w3-demo get pvc nginx-data
NAME            STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
nginx-data      Bound    csi-082c7776-ed2e-4301-80e1-fd4d7cf02fe2   50Gi        RWO             oci-bv          <unset>                  11m
PS E:\work\Ejada\terraform\nonprd\week3-containerized-oke>
```

**C02** — PVC `nginx-data` Bound, 50Gi, StorageClass `oci-bv`.

### C03 — block volume

| Block volumes                                                                                                                          |           |       |                     |                        |                          |                     |               |                            |     |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|---------------------|------------------------|--------------------------|---------------------|---------------|----------------------------|-----|
| Block volumes provide high-performance network storage to support a broad range of I/O intensive workloads. <a href="#">Learn more</a> |           |       |                     |                        |                          |                     |               |                            |     |
| <div> <input type="text" value="Search and Filter"/> <input type="button" value="Search"/> </div>                                      |           |       |                     |                        |                          |                     |               |                            |     |
| Applied filters: <span>Compartment: intern-18-salah-abdelhady-cmp</span>                                                               |           |       |                     |                        |                          |                     |               |                            |     |
| <input type="button" value="Create block volume"/>                                                                                     |           |       |                     |                        |                          |                     |               |                            |     |
| Name                                                                                                                                   | State     | Size  | Default performance | Auto-tuned performance | Cross region replication | Availability domain | Backup policy | Created                    |     |
| csi-082c7776-ed2e-4301-80e1-fd4d7cf02fe2                                                                                               | Available | 50 GB | VPU: 10             | —                      | Off                      | oXVME-JEDDAH-1-AD-1 | —             | Aug 15, 2026, 21:56:38 UTC | ... |
| Page 1 of 1 (1 - 1 of 1 total items)                                                                                                   |           |       |                     |                        |                          |                     |               |                            |     |
| Items per page: 25                                                                                                                     |           |       |                     |                        |                          |                     |               |                            |     |

**C03** — Block volumes list: the CSI-provisioned volume, 50 GB, Available.

| Block volume information     |                                                                                                                                                | Auto-tuned performance policies                                                                                                                                                                                                        |                                                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Availability domain          | oXVME-JEDDAH-1-AD-1                                                                                                                            | Performance based auto-tune                                                                                                                                                                                                            | Off<br>Enables automatic performance level adjustments in real-time, based on the volume workload requirements. <a href="#">Learn more</a> |
| Cluster placement group      | None                                                                                                                                           | Detached volume auto-tune                                                                                                                                                                                                              | Off<br>When enabled, volume performance is adjusted to the lower cost level when volume is detached. <a href="#">Learn more</a>            |
| Cluster placement group OCID | None                                                                                                                                           | Cross region replication                                                                                                                                                                                                               |                                                                                                                                            |
| Compartment                  | ociejada (root)/2026-intern-program/intern-18-salah-abdelhady-cmp                                                                              |                                                                                                                                                                                                                                        |                                                                                                                                            |
| OCID                         | ...abvgkljg4xbk3gt4rglkg5nspcn2k35bqeff45gxxumlgnd4fyxxmq <input type="button" value="Copy"/>                                                  | Managed by                                                                                                                                                                                                                             |                                                                                                                                            |
| Created                      | Aug 15, 2026, 21:56:38 UTC                                                                                                                     |                                                                                                                                                                                                                                        |                                                                                                                                            |
| Size                         | 50 GB<br><small>Resizing a volume is available only in certain situations. See <a href="#">resizing a volume</a> for more information.</small> | Volume<br><small>If the volume is part of a volume group, then the group manages the cross region replication setting. You must change the cross region replication setting for the group or remove the volume from the group.</small> |                                                                                                                                            |

**C03b** — CSI volume detail (`csi-082c7776-...`), 50 GB, in the `intern-18` compartment.



**C04 — Service external IP**

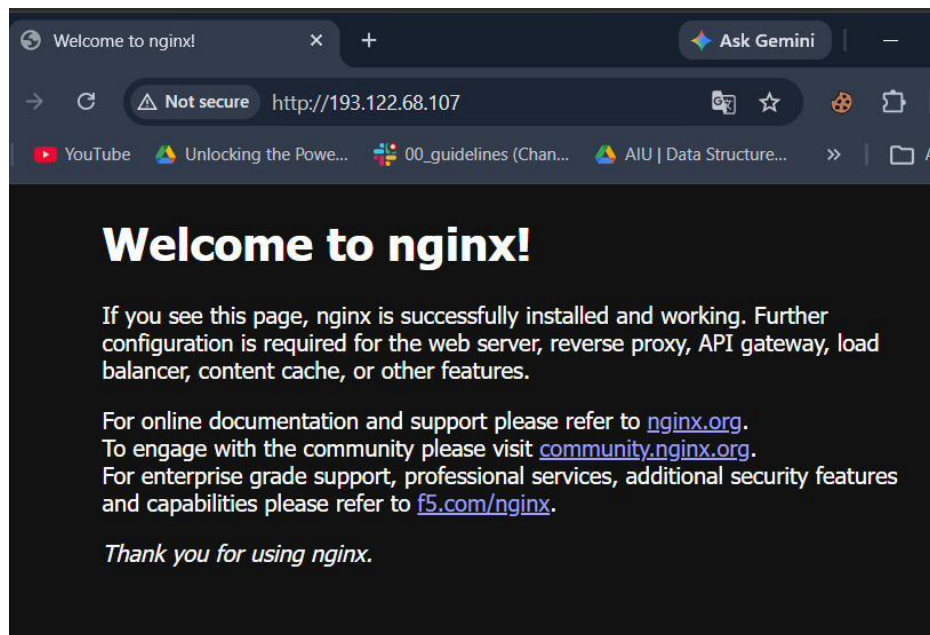
```
PS E:\work\Ejada\terraform\nonprd\week3-containerized-oke> kubectl -n w3-demo get svc nginx
NAME      TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)        AGE
nginx     LoadBalancer 10.96.15.207   193.122.68.107 80:31785/TCP    13m
PS E:\work\Ejada\terraform\nonprd\week3-containerized-oke>
```

**C04** — `kubectl -n w3-demo get svc nginx` — external IP 193.122.68.107.

**C05 — console load balancer**

The screenshot shows the OCI Load Balancers console. At the top, there's a search bar and a filter for 'Compartment: intern-18-salah-abdelhady-ome'. Below this is a 'Create load balancer' button. The main table lists the load balancers with columns: Name, State, IP address, Shape, Overall health, and Created. One load balancer is listed with Name '230de148-dc3d-4d1c-b71d-de45179019e8', State 'Active', IP address '193.122.68.107 (public)', Shape 'Flexible', Overall health 'OK', and Created 'Aug 15, 2026, 21:56:10 UTC'. The bottom of the console shows 'Page 1 of 1 (1 - 1 of 1 total items)' and 'Items per page 25'.

**C05** — Load balancers list: the Flexible load balancer, public IP 193.122.68.107, health Ok.

**C06 — browser check**

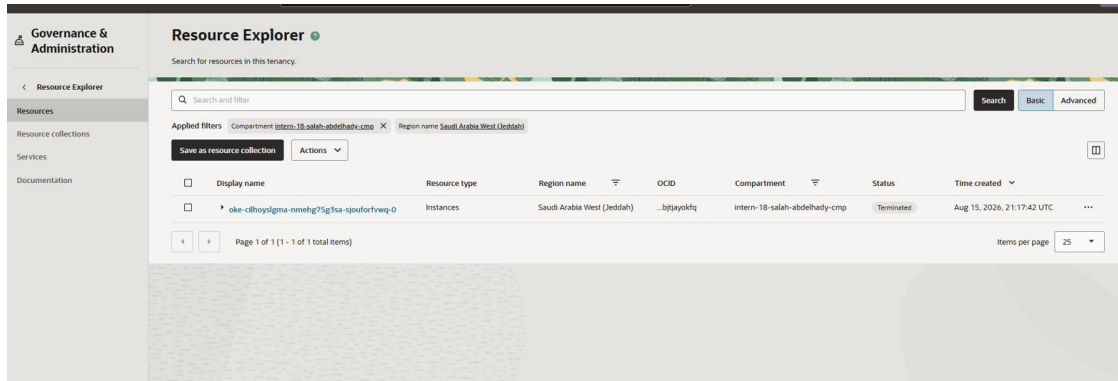
**C06** — Browser at `http://193.122.68.107/` showing the nginx welcome page.

**C07 — volume mount**

```
nginx LoadBalancer 10.96.15.207 193.122.68.107 80:31785/TCP 13m
PS E:\work\Ejada\terraform\nonprd\week3-containerized-oke> kubectl -n w3-demo exec deploy/nginx -- df -h /data
Filesystem      Size      Used Available Use% Mounted on
/dev/sdb        48.9G     24.0K    48.9G    0% /data
PS E:\work\Ejada\terraform\nonprd\week3-containerized-oke>
```

**C07** — `kubectl exec ... -- df -h /data` — /dev/sdb mounted on /data.

## C08 — post-destroy inventory



**C08** — Resource Explorer after destroy: only a TERMINATED OKE worker ghost remains.

## 10. Issues and Troubleshooting

Both issues below were resolved before the destroy step, and neither left an open follow-up: the stack was destroyed cleanly once evidence capture was complete.

### 10.1 Cluster creation was rejected with a 404 authorization error

#### Observation

The first apply with `enable_oke = true` failed on the cluster create call, returning 404-NotAuthorizedOrNotFound. The node pool was never attempted because Terraform stopped at the failed resource.

```
Error: 404-NotAuthorizedOrNotFound, Authorization failed or requested resource not found
on oke.tf line 1, in resource "oci_containerengine_cluster" "this":
  1: resource "oci_containerengine_cluster" "this" {
```

#### Investigation

- Re-checked the compartment target in the error output against `intern-18-salah-abdelhady-cmp` — the compartment itself was correct, ruling out a simple typo or wrong-OCID mistake.
- Re-ran terraform plan against the existing state to confirm the failure was isolated to the cluster create call and not a wider provider or network issue — the network module's resources (VCN, gateways, subnets) all planned clean with no changes.
- Reviewed the intern IAM policy statements available at the time and found no explicit grant for the cluster-family resource family or the OKE service, which narrowed the cause to a missing policy statement rather than a Terraform configuration defect.

#### Resolution

Requested a policy update from the mentoring team to add create permission on cluster-family for the OKE service in the `intern-18` compartment. No changes were needed on the Terraform side — the module and variables were already correct for the intended cluster shape.

#### Verification

Once the policy statement was in place, terraform apply was re-run unchanged and completed in full: cluster `w3-oke` reached ACTIVE in roughly 8m40s and node pool `w3-np` followed in roughly 5m26s. terraform state list and the OCI CLI snapshot in Section 8.1 confirm both resources as ACTIVE with CNI `OCI_VCN_IP_NATIVE`.

### 10.2 Reapplying the storage class conflicted with the cluster's built-in definition

#### Observation

`kubectl apply -f k8s/storageclass.yaml` was rejected because OKE's control plane already provisions the `oci-bv` StorageClass at cluster creation, and the manifest's apply attempted to update fields the API treats as immutable on an existing StorageClass.

#### Investigation

- Ran `kubectl get storageclass` and confirmed `oci-bv` already existed with the same provisioner (`blockvolume.csi.oraclecloud.com`) that the manifest declared.
- Compared the manifest's parameters against the live object and confirmed the intent was identical — the manifest was documenting the class the cluster already exposes, not introducing a new one.

### Resolution

Skipped the redundant apply of storageclass.yaml and proceeded directly to pvc.yaml, which references storageClassName: oci-bv by name. The manifest remains in the repository as the explicit, version-controlled definition of the class the workload depends on.

### Verification

kubectl -n w3-demo get pvc nginx-data showed Bound against the existing oci-bv class within seconds of applying the PVC, and the volume mounted successfully inside the pod (Section 7.4, Screenshot C02 and C07).

## 11. Cleanup and Destroy Verification

1. Validated pods, PVC binding, Service external IP, the browser check, and the /data mount while the stack was live.
2. Deleted the Kubernetes Service, Deployment, PVC, and namespace.
3. Ran terraform destroy -auto-approve, which completed with 36 resources destroyed at exit code 0.
4. Confirmed terraform state list returned empty.
5. Ran an OCI CLI post-destroy inventory across me-jeddah-1 for the intern-18 compartment.

| Family                                                   | Residual live?                                    |
|----------------------------------------------------------|---------------------------------------------------|
| VCN / subnets / Internet Gateway / NAT / Service Gateway | No                                                |
| NSGs / route tables / security lists                     | No                                                |
| OKE clusters                                             | No (DELETED list-API ghost only)                  |
| Load balancers                                           | No                                                |
| Block / boot volumes                                     | No (TERMINATED CSI and boot ghosts only)          |
| Instances                                                | No RUNNING instances — TERMINATED OKE worker only |
| Log groups                                               | No                                                |

The compartment is lab-empty for billable Week 3 resources. The intern-18 compartment object itself was not deleted, consistent with the shared, long-lived nature of the intern compartment across weekly labs. Console confirmation: Screenshot C08 (Section 9.3).

## 12. Document Index

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This report is the complete Week 3 Lab 3 submission package. It contains, in one file:

- Sections 1–6 — objectives, environment, architecture, source inventory, variables, and design standards, with Figures 1–5.
- Sections 7–8 — the full command history for the successful path and the evidence catalogue backing it.
- Section 9 — the curated Console and kubectl screenshot set (31 captures) organized by stage.
- Sections 10–11 — the two issues encountered and how each was diagnosed and resolved, and the destroy verification.

Working stack (source-controlled): [github.com/Salahmohameds/ejada-oci-week1-terraform](https://github.com/Salahmohameds/ejada-oci-week1-terraform)

*End of Terraform documentation for Week 3 Lab 3 — covering code, commands, evidence, diagrams, and the destroy.*